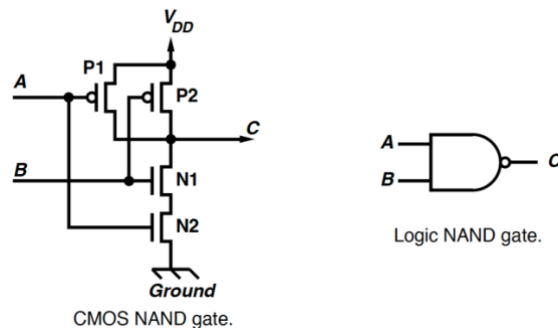


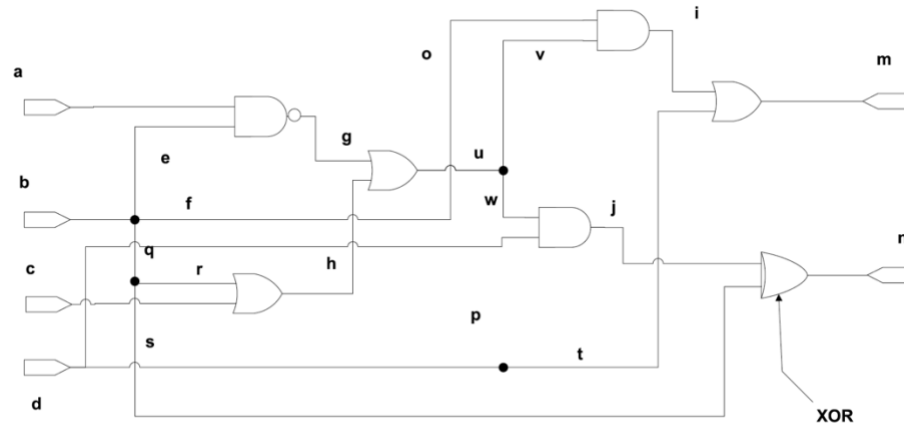
Homework # 3

1. For a two-input CMOS NAND circuit:



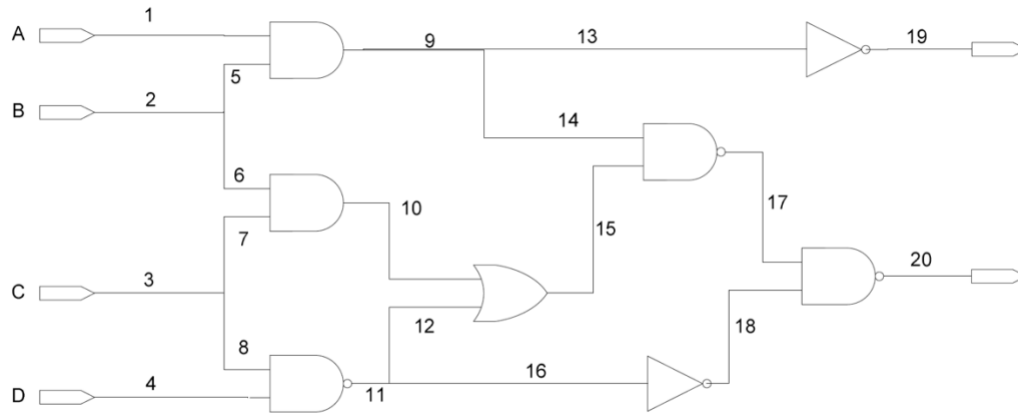
- Find a two-pattern test for each single-transistor stuck-open fault
- Rearrange the eight vectors in a compact set so the sequence is as short as possible
- For each stuck-at fault of the NAND gate, find an equivalent transistor (stuck-open, stuck-short, or a combination) fault.

2. Show that the two faults a SA0 and c SA1 are equivalent in the circuit below.



3. For this problem, use the table in the figure in the next page.

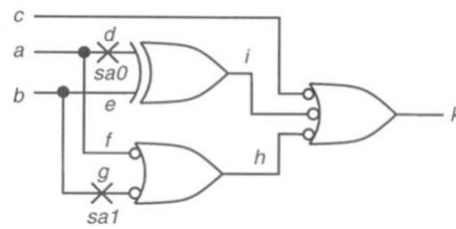
- Table 1 gives a fault list for the circuit, after equivalence fault collapsing. Fill in the blanks with equivalent fault sets for each listed fault. (Validity check: the total number of faults in table should be 40)



Fault	List all equivalent Faults
1/1	$\emptyset$
9/0	
9/1	
5/1	
2/0	
2/1	
10/0	
6/1	
7/1	
3/0	
3/1	
11/0	
11/1	
8/1	
4/1	
17/1	
14/1	
15/1	10/1, 12/1
12/0	
18/1	
19/0	
19/1	
20/0	
20/1	

Table 1

4. For the circuit below:



- Enumerate the minimal set of single stuck-at-faults that must be tested according to the Checkpoint Theorem.
- Show that the two faults d SA0 and g SA1 are equivalent.

Note: Use the following definition for checkpoints: *Checkpoints : Primary inputs and fanout branches of a combinational circuit consisting only of BOOLEAN gates are called the checkpoints.* Thus, expand XOR gate as in BA Figure 4.9, which is same as the one shown below.